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RESEARCH ARTICLE

A Multi-Market Data-Driven Stock Price Prediction System: Model Optimization and Empirical Study

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ABSTRACT As global financial markets become increasingly interconnected, relying solely on historical data from a single market is insufficient for achieving high prediction accuracy in stock price forecasting. It is essential to consider the impact of cross-market transactions on China's stock market fluctuations. This study investigates this impact by constructing a comprehensive dataset that includes stock market, bond market, and foreign exchange market data from multiple countries. A deep learning algorithm, combined with the grid search method, was employed to compare different structural hyperparameters of four models: LSTM, GRU, MLP, and Transformer. The results indicate that the GRU model outperformed the others in terms of prediction accuracy. Consequently, a GRU-Arbitrage trading system based on optimal structural hyperparameters was developed. This system effectively integrates international and domestic information for stock price prediction, providing investors with a practical and accurate tool. The research findings show that the mean squared error (MSE) of stock prices predicted using international market transaction information for the Chinese stock market is 0.0069, followed by 0.0035 for the bond market and 0.0102 for the foreign exchange market. These results verify the significant impact of cross-market data on predicting China's stock market fluctuations.

INDEX TERMS Stock index prediction, cross-market, hyperparameter tuning, GRU trading system.

I. INTRODUCTION

In stock market forecasts, many types of information reflect prices. Besides technical indicators and fundamental data, macroeconomic variables related to the target market, such as exchange rates, bond yields, and oil prices, also play a significant role [1], [2], [3]. As China accelerates its opening up and economic and financial activities become more frequent, stock forecasting faces the necessity and challenge of understanding the coupling relationships (cross-market behaviors) between different market behaviors [4], [5]. These complex interactions amplify the influence of cross-market macroeconomic variables on the stock market. However, the difficulty of establishing multi-market relationships through statistical models has hindered research progress in this area.

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The development of deep learning technology has made it easier for researchers to establish connections between multiple markets, enhancing the potential for more accurate stock market predictions.

Research on multi-market behavior should further consider the linkage effects of global financial markets and integrate data from more countries and regions to build a comprehensive prediction model. For instance, Scholz et al. [12] incorporated bond market data to forecast the Danish stock market, effectively improving stock returns. Similarly, Neely Ch and Higbee [13] utilized cross-market exchange rate data, including DEM, JPY, CHF, and GBP against the U.S. dollar in four foreign exchange markets, to study the U.S. stock market. Their findings indicate that foreign exchange data significantly predicts stock returns. This demonstrates the profitability of combining technical indicators with cross-market variables for stock market predictions. However, relying solely on a single cross-market variable to improve the prediction accuracy of the Chinese stock market may be due to the special nature of the data, and it is impossible to verify the accuracy of cross-market data continuing to be used as labels for stock predictions.

Recent research on the application of deep learning technology in stock prediction shows that deep learning can extract features from datasets to predict stock prices more accurately [14], [15]. Among these, GRU [16], ANN [17], LSTM [18], and Transformer [19] have received more attention, along with research on improvements to these models. Further research indicates that the performance of these algorithm models is significantly influenced by the optimization of hyperparameters [20]. However, the relationship between model parameter tuning and algorithm performance has been sparsely discussed in the existing literature. Improper hyperparameter settings can affect the reproducibility of research results [21]. Careless hyperparameter settings, a key factor affecting the performance of deep learning models, may lead to worse prediction results [22], [23]. Moreover, current stock prediction research tends to focus on the model level and often overlooks performance in actual trading scenarios [24], [25]. Therefore, future research should place greater emphasis on hyperparameter optimization and enhance the robustness and effectiveness of models in practical applications.

This paper establishes a model-tuning framework utilizing the grid search method to select the optimal model for cross-market datasets, develops a cross-market GRU-Arbitrage trading system, and compares its trading results with those of three other models. Investment performance is evaluated using the Sharpe ratio, return, and risk financial indicators.

The primary contributions of this study are as follows: (1) Constructing cross-national stock, exchange rate, and bond data as labeled data to predict Chinese stock market prices, thereby broadening the sources of labeled data and compensating for the influence of international market fluctuations in stock market prediction; (2) Identifying the optimal structural hyperparameters for deep learning models while considering economic heterogeneity, and demonstrating that the GRU model's MSE is at least 0.003 better than that of the other three models, highlighting the importance of structural hyperparameters as optimization parameters in financial datasets; (3) Developing a GRU-Arbitrage algorithmic trading system and conducting a comparative analysis of its investment performance against three other deep learning algorithms, revealing that the GRU-Arbitrage trading system's Sharpe ratio is at least 0.06 higher than that of the other systems in the studied cross-market dataset.

TABLE 1. Empirical studies on cross-market data from three investment targets data sets in stock prediction research.

Article	Sample	Period	label	target
[6]	SSEC, S&P 500, FTSE 100, Nikkei 225, DAX, CAC 40, FTSE MIB, and S&P/TSX	Jan 2000 to Dec 2016	5-minute sampling	Volatility of the Chinese Stock Market
[7]	SSC, S&P 500, N225, FTSE 100, CAC40, and DAX	Jan 2004 to Apr 2022	Daily logarithmic return and Daily realized volatility	Volatility of the Chinese Stock Market
[8]	CNY, JPY, KRW per USD, Shanghai Composite, Nikkei 225, and KOSPI	Jul 2005 to Sep 2013	Closing prices	Causality relationships between foreign exchange and stock markets
[9]	RMB to USD, Shanghai Composite Index (SZ), and S&P 500	Jan 2011 to Nov 2019	Natural logarithm	Spillover effect
[10]	S&P 500, Nikkei 225, DAX, CAC40, FTSE-100, Italian Index, Shanghai Composite Index, Australian S&P 200, and Toronto 300, 10-year bond daily returns of these countries	Jan 2016 to Feb 2017	Returns	Correlation coefficients
[11]	10-year Colombian bond yield, Colombian stock index, credit default swap (CDS) spreads on 5-year Colombian bonds in US dollars, COP/USD exchange rate and its 10-year expected value, and 10-year US Treasury yield	Jun 2004 to Nov 2013	10-year Colombian bond yield, CDS spreads expressed as a percentage, and the logarithm of others	US market volatility, US Treasury term premium, and 10-year US Treasury yield

The structure of this paper is as follows: Section II reviews recent related work; Section III outlines the research methodology; Section IV presents experimental procedures and results; Section V provides a summary of the study and discusses future research directions.

II. RELATED WORK

In the ensuing discussion, this paper will explore research related to cross-market data, deep learning models, and trading systems within the context of the stock market.

A. CROSS-MARKET DATA

Currently, most research on the stock market relies on the historical information of stocks themselves to predict future trends, as historical data encapsulates all information regarding stock price fluctuations [26], [27]. However, with global economic integration, connections between regional financial markets have become closer. Many economists have identified interdependencies among data from different markets [28], [29], [30].

Table 1 presents recent studies on the cross-market contagion effect on stock markets. Peng et al. [6] used seven indices, including the S&P 500, FTSE 100, and Nikkei 225, to represent their respective national stock markets and examined the influence of international stock market volatility on fluctuations in the Chinese stock market. The study found that stock market fluctuations in these seven countries significantly contribute to predicting future fluctuations in the Chinese stock market. Wang et al. [7] analyzed the daily log returns and daily realized volatility derived from five national stock indices, providing insights into the volatility of the Chinese stock market from different perspectives. Their findings confirm the explanatory and predictive roles of cross-market stock information on the Chinese stock market.

In addition to the mutual influence of international stock markets in predicting volatility, cross-asset markets such as bonds and foreign exchange also affect stock markets. For instance, Park et al. [8] examined the relationship between the stock markets in China, Japan, and South Korea and the foreign exchange market. The study revealed significant correlations between stock markets and foreign exchange markets in Japan and South Korea, but no significant correlation was found in China. Although it is uncertain whether the correlation between the stock market and the foreign exchange market is related to the level of development, the gradual opening of Chinese financial markets has increased the correlation between the Chinese stock market and the foreign exchange market. Su [9] used data from the yuan-dollar foreign exchange market and the U.S. stock market to examine the relationship with the Chinese stock market during the period from 2011 to 2019, finding significant volatility spillover effects from the Chinese stock market to both the U.S. stock market and the Renminbi foreign exchange market. Chen et al. [10] conducted a statistical analysis on the correlation between nine major national stock markets and bond markets, revealing significant negative correlations between each country's stock market and bond market. Guarín et al. [11] analyzed the volatility of US 10-year Treasury bonds using cross-market data, demonstrating significant associations between US Treasury bonds and Colombian stocks, bonds, and currencies across various periods.

Therefore, this study aims to establish a dataset encompassing cross-national stock markets, bond markets, and foreign exchange markets to predict prices in the Chinese stock market.

TABLE 2. Current studies of deep learning algorithms in stock price prediction.

Article	Benchmarks	Model	Tuning Techniques
[35]	SVM, LSTM, MLP, RNN, Naive Bayes, Decision Tree	MLP, RNN, LSTM	Not mentioned
[36]	MLP, LSTM	MLP, LSTM	Not mentioned
[34]	GRU, LSTM, LSTM-GRU, ARIMA	LSTM-GRU	grid search
[33]	GRU, LSTM, TGRU	TGRU	Not mentioned
[32]	LSTM	LSTM	Genetic algorithm
[37]	RNN, CNN, Transformer, LSTM	Transformer	Results analysis (Epoch only)
[19]	Transformer	Transformer	Results analysis (Epoch only)

B. DEEP LEARNING MODEL

In addition to research on cross-market data, using deep learning models to predict stock prices is also one of the hot research directions. Table 2 summarizes the research status of deep learning algorithms applied in the field of stock prediction. Kumbure et al. [31] reviewed extensive literature and reported a substantial increase in research employing deep learning models for stock forecasting over the past five years. For instance, Chung and Shin [32] integrated LSTM networks with genetic algorithms to optimize the LSTM topology and hyperparameters, thereby enhancing predictive accuracy in stock markets. Similarly, Minh et al. [33] refined the structure of the Two-Stream Gated Recurrent Unit (TGRU) model to bolster its predictive performance. Their study conducted a comparative analysis of the accuracy, precision, and recall of GRU and LSTM models in forecasting stock data, revealing that the TGRU model demonstrated superior performance. Ling and Belaidan [34] used four models to predict individual stocks and determined that the LSTM-GRU hybrid model exhibited the highest performance across various metrics, including MSE, RMSE, MAE, and F-score.

Secondly, research related to MLP models is also notable. Li and Liao [35] utilized Support Vector Machines, Naive Bayes, and MLP models to analyze data from a Chinese listed company (code 000592) and found that MLP, RNN, and LSTM models significantly outperformed other models in terms of accuracy. Similarly, Labiad et al. [36] examined trends in three stocks with varying market values and aimed to determine the optimal model structure by fine-tuning the hyperparameters of MLP and LSTM models, though specific tuning techniques were not disclosed. The results demonstrated that both MLP and LSTM models performed effectively in stock prediction.

Finally, the Transformer model has gained considerable attention. Muhammad et al. [19] forecasted daily and weekly closing prices for eight distinct stocks, demonstrating that the Transformer model produced convergent results with prediction errors, measured by MSE and RMSE, ranging from 0.1 to 0.01 on the test set. These results indicate robust performance in predicting stock closing prices; however, comparisons with other models were not included. Wang et al. [37] compared Transformer models with RNN, CNN, and LSTM models in forecasting four major international indices. Their findings revealed that the Transformer model achieved superior predictive accuracy across metrics such as MAE, MSE, and MAPE, though the study only addressed Epoch hyperparameter optimization. Consequently, this study aims to employ a cross-market dataset to train MLP, LSTM, GRU, and Transformer deep learning models for predict Chinese stock prices. The research will focus on evaluating the impact of model structural hyperparameters on prediction performance.

C. TRADE SYSTEM

The strategies incorporated into a trading system are also critical. Brzeszczynski and Ibrahim [38] developed a trading system utilizing foreign exchange data that integrated fuzzy logic-based strategies. This approach combined external information, represented by random parameters, with domestic momentum indicators such as the Relative Strength Index (RSI) to generate trading signals. While the analysis indicated potential profit opportunities, the study noted that these opportunities are challenging to realize in actual trading scenarios. Han et al. [39] investigated the determination of buy, sell, and hold signals through improvements in labeling methods. Their analysis, which involved evaluating metrics such as winning rate, return rate, and profit coefficient, revealed that enhancing prediction accuracy can improve the trading system's winning rate. However, it does not necessarily lead to a proportional increase in the system's overall return.

Thus, optimizing strategies to achieve higher returns remains a central focus in trading system research. Dymova et al. [2] employed the buy-and-hold strategy as a benchmark to compare total returns. Their study demonstrated that a trading strategy combining fuzzy logic with Dempster-Shafer evidence theory yielded a total return 150% higher than the benchmark. Similarly, Lohrmann and Luukka [40] developed four strategies to examine trading system rules, using the classic buy-and-hold strategy as the benchmark. Their evaluation of daily average returns revealed that the performance of various strategies is influenced by transaction costs. Building on this, the present study aims to develop a trading system tailored for cross-market data. It will assess the system's effectiveness and potential application value in the Chinese stock market by considering the investment efficiency of the benchmark strategy as well as practical factors such as transaction fees.

III. PROPOSED TRADING SYSTEM

This study focuses on three key aspects: constructing the dataset, predicting prices, and developing a trading system. The initial step involves creating a cross-market dataset that encompasses stocks, bonds, and foreign exchange. In the second step, deep learning models known for their efficacy in forecasting financial time series data are selected. These models are then trained on the cross-market dataset, with parameters optimized to minimize the MSE. The final step involves constructing a trading system using the optimal model to execute investment strategies. Investment performance is evaluated using metrics such as daily average return, risk rate, and Sharpe ratio. Figure 1 depicts the three components of the trading system developed in this study.

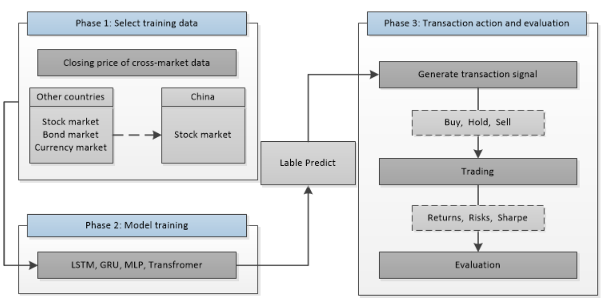


FIGURE 1. The framework of the proposed model.

A. PHASE 1: SELECT TRAINING DATA

This study gathered cross-market trading data for stocks, bonds, and currencies from January 2014 to January 2024. Detailed descriptions of the asset data are provided in Section IV-A. The collected data exhibited inconsistencies in trading times and mismatched lengths across different assets. To address these issues, the collected data were arranged by time index, retaining only observations that

TABLE 3. Comparison of mean MSE values for different window lengths T_w.

T_w	stock	bond	Currency Rate
15	0.028	0.026	0.122
20	0.008	0.007	0.012
25	0.034	0.011	0.03
30	0.017	0.012	0.015
35	0.029	0.008	0.041
40	0.012	0.02	0.0146
45	0.01	0.014	0.016
50	0.04	0.013	0.019

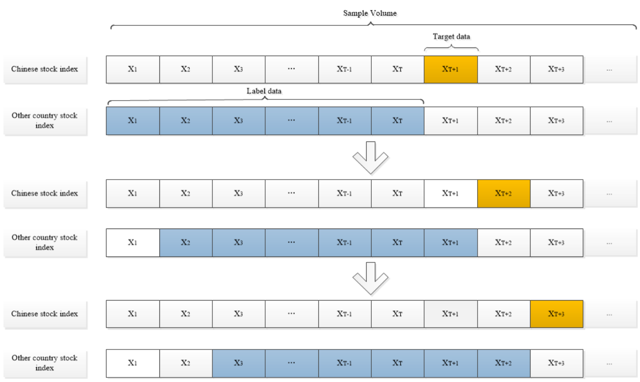


FIGURE 2. Cross-national stock dataset segmented using a rolling window method: Blue represents label data, orange represents target data.

aligned with the trading days of the SZ399001 index while removing records from other markets on non-trading days to ensure data consistency and prevent information leakage. Additionally, variations in trading schedules across markets were standardized, as certain international markets remained open on Chinese public holidays while the Chinese market was closed.

The absolute values of price fluctuations across different assets vary significantly, potentially compromising the stability of the model's predictive performance when evaluated using these absolute values. To ensure an accurate assessment of the model's prediction accuracy across diverse datasets, it is essential to normalize the data. The normalization procedure is outlined as follows:

$$z = \frac{x - \mu}{\sigma} \tag{1}$$

In this context, z represents the normalized data, x denotes the underlying data, μ and σ refer to the sample mean and standard deviation of the underlying, respectively.

Normalizing the data mitigates volatility, thereby improving the model's convergence. Post-normalization, it is also crucial to label the data and define targets to ensure that the model's fitting process aligns with the anticipated objectives.

In this study, data spanning T_w time periods are employed as labels, while the data from the subsequent day serve as targets. To determine the optimal label length, we conducted preliminary experiments on three datasets, each selecting one subset of data for evaluation. Without considering model parameters, we tested different window lengths from 15 to 50 days and recorded the mean MSE values for four models. The detailed results of the preliminary validation of each model for T_w can be found in Appendix B. The results, as shown in Table 3, indicate that T_w=20 days yields the lowest MSE and the best predictive performance. Therefore, we set T_w=20 as the default value for further hyperparameter optimization.

B. PHASE 2: MODEL TRAINING

This study utilizes four widely adopted deep learning models for forecasting the Chinese stock market. The selected models include LSTM, GRU, MLP, and Transformer.

1) MODEL

Long Short-Term Memory (LSTM) networks are an enhanced version of RNNs, specifically designed to overcome the challenges of gradient explosion and gradient vanishing encountered during the training of extended sequences [41]. Unlike traditional RNNs, LSTM incorporate specialized units for managing information flow, mitigating issues related to long-term dependencies in sequence data. The inherent functionality of LSTM networks facilitates the retention of information across prolonged sequences, making them particularly effective for addressing long-sequence problems [42].

Gated Recurrent Units (GRU) are a class of recurrent neural network models [43] frequently used for time series forecasting. GRU effectively address gradient vanishing and explosion issues due to their distinctive architecture and mechanisms within the backpropagation algorithm. This design enhances the model's capacity to capture and propagate sequential information while improving gradient stability. As a streamlined alternative to LSTM networks, GRU offer increased computational efficiency and superior processing of input sequences while retaining long-term memory [44].

The Multilayer Perceptron (MLP) is a feedforward artificial neural network model [45]. It features a relatively straightforward architecture comprising an input layer, one or more hidden layers, and an output layer. It is designed to fit and compute input data effectively and, in theory, can approximate any functional form. The activation function plays a crucial role in enabling its nonlinear learning capabilities.

In contrast, the Transformer model utilizes an encoder-decoder architecture and entirely eschews recurrent structures, relying instead on an attention mechanism to manage global dependencies between inputs and outputs. This architecture is particularly adept at handling long sequence data, as the self-attention mechanism facilitates the retention of long-range dependencies within the data. Consequently, the

Algorithm 1 Trading Strategy Based on Predicted Prices

Require: T: Label length, *trade_end*: Stop trading date, *threshold*: Percent threshold

for t to T **do**
    $prediction[t+1] \leftarrow model.predict(close\_price[t-T:t])$
    $trade\_threshold \leftarrow close\_price[t] \times threshold$ ▷ Calculate trade threshold
    if $prediction[t+1] > close\_price[t] + trade\_threshold$ **then**
        `cover(); buy();`
    else if $prediction[t+1] < close\_price[t] - trade\_threshold$ **then**
        `sell(); short();`
    end if
    if `date == trade_end` **then**
        break ▷ Terminate trading
    end if
end for
Calculate:

$$SR = \frac{R_p - R_f}{\sigma_p} \cdot \sqrt{240},\ R_p = \frac{\sum_{t=1}^{T} R_t}{T}, \quad \sigma_p = \sqrt{\frac{\sum_{t=1}^{T}(R_t - \bar{R})^2}{T}}$$

▷ Analyze investment performance

return SR, R_p, σ_p

Transformer model can process data in parallel, significantly enhancing training efficiency [46].

2) LOSS FUNCTION

The loss function used in this study is the Mean Squared Error (MSE), which is utilized to assess the predictive performance of the models. The MSE quantifies the average squared difference between predicted and actual values, providing a precise measure of the absolute prediction error and enhancing the understanding of discrepancies between the predicted and observed outcomes [47]. The formulation of the loss function is outlined as follows:

$$MSE = \frac{1}{N}\sum_{i=1}^{N}(y_{\text{true}_i} - y_{\text{pred}_i})^2 \tag{2}$$

where y_{pred_i} represents the model's predicted value, y_{true_i} represents the actual value, and N denotes the total number of samples, with y_{pred_i} and y_{true_i} being values normalized during computation.

C. PHASE 3: TRANSACTION ACTION AND SIMULATION

Utilize the loss function to evaluate the predictive performance of the four deep learning models on the test set. Identify the model that demonstrates the optimal performance according to the loss function and develop a trading system based on the established trading rules. Ensure that the trading rules exhibit superior investment efficiency. Subsequently, perform backtesting to assess the investment performance, calculating metrics such as the daily average return rate, risk rate, and Sharpe ratio. Conduct a comparative analysis of the investment performance across the three remaining model-based trading systems to confirm that the GRU trading system delivers the most effective investment results.

1) TRADING STRATEGY

This paper constructs a GRU-based arbitrage strategy for algorithmic trading on cross-market datasets. The trading rules are detailed in Algorithm 1. First, prepare the verification dataset and determine the label length required for transaction backtesting, as well as the date parameters for ceasing trading. Next, use the deep learning model to input the label data, `close_price[t-T:t]`, of length T, to predict the following day's closing price, `price_prediction[t+1]`. The trading signal is determined by comparing `price_prediction[t+1]` with today's closing price (i.e., threshold = 0): if tomorrow's closing price is predicted to be higher than today's, execute the `cover()` and `buy()` operations. Conversely, if tomorrow's price is predicted to be lower, perform the `sell()` and `short()` operations. Finally, terminate the transactions upon reaching the specified deadline. Calculate the daily average return, risk rate, and Sharpe ratio, and then close the system.

By comparing the system's trading rules with the classic buy-and-hold strategy, the investment efficiency of the trading system is ensured. In the data processing phase, to adjust transaction frequency and control transaction costs, the system's trading signal conditions are optimized by setting threshold values to 0%, 0.05%, 0.1%, and 0.5%. We consider 0% as the optimal threshold, as it provides the best trading solution. The values of 0.05% and 0.1% are mainly used for comparison with 0%, to observe the effect of increasing the threshold on the trading system. The 0.5% threshold is considered as a boundary value. Although its performance is not as effective as the previous thresholds, it is included in the experiment to provide a comprehensive comparison of the impact of different thresholds on the performance of the trading system. In this study, a deep learning algorithm is utilized to manage the arbitrage strategy, enhancing prediction accuracy and maximizing returns.

2) PERFORMANCE EVALUATION

To evaluate investment performance, appropriate metrics must be selected. This article employs the Sharpe ratio to measure the returns generated by a strategy relative to the risk undertaken. The Sharpe ratio is defined as follows:

$$SR = \frac{R_p - R_f}{\sigma_p} \cdot \sqrt{240} \tag{3}$$

In this context, R_p represents the daily average return rate of the investment portfolio (assets), while R_f denotes the daily average risk-free rate. The risk-free rate is chosen to be 0.015, based on the one-year deposit rate of the People's Bank of China, due to its relevance to the Chinese market, stability, and widespread applicability. As the benchmark rate set by the central bank, it reflects the overall economic conditions in China and represents the risk-free return for ordinary investors. Therefore, with represented as $0.015/\sqrt{240}$. σ_p represents the daily average volatility (standard deviation) of the investment portfolio (assets).

The daily profits from system trading can be used to calculate the daily average return rate R_f, which is essential for computing the Sharpe ratio. The detailed calculation formula is as follows:

$$R_p = \frac{\sum_{t=1}^{T} R_t}{T} \tag{4}$$

where R_t represents the profit generated from system trading on each trading day, with a total of T trading days.

The standard deviation of returns, also known as daily average volatility, quantifies the fluctuation in returns. This study employs this metric to measure the strategy's risk. The specific formula for calculating the daily average volatility σ_p is as follows:

$$\sigma_p = \sqrt{\frac{\sum_{t=1}^{T} (R_t - \bar{R})^2}{T}} \tag{5}$$

where $\bar{R}$ represents the average return, and T denotes the total number of trading days in a year. In this study, it is assumed that there are 240 trading days in a year.

IV. EXPERIMENTAL STEP AND RESULTS

This study is conducted in three steps: (a) Collect cross-market data for stocks, bonds, and foreign exchange; (b) Perform hyperparameter tuning for four deep learning algorithms (MLP, LSTM, GRU, and Transformer) and select the optimal algorithm for predicting the Chinese stock market index (SZ399001); (c) Construct a trading system based on the best algorithm and perform a horizontal comparison of the investment performance with trading systems using other deep learning algorithms.

A. DATASET

This study's data primarily consists of two parts: first, international major stock market indices from 15 regions obtained from Tushare (https://tushare.pro); second, corresponding data from the Wind database, including government-issued 10-year bond yield indices and foreign exchange rates against the Chinese Yuan. Compared to the target stock index SZ399001, the collected stock market dataset is characterized by the single feature of "cross-national." In contrast, the bond market and currency rate market datasets exhibit dual features: "cross-national" and "cross-asset."

TABLE 4. Data codes for various market.

Nation	Stock Market	Bond Market*	Currency Rate Market
America	RUT,SPX, IXIC,DJI	G0000891	M0000185
Australia	AS51	G0006860	M0068015
Brazil	IBOVESPA	G1924535	M0161089
Canada	SPTSX	G1800591	M0161088
China	SZ399001**	×	×
England	FTSE	×	M0000189
EU	CSX5P	×	M0000186
France	FCHI	G1400003	×
Germany	GDAXI	B2559386	×
Hong Kong	HSI	G8322969	M0000187
India	SENSEX	G0266641	M0161072
Japan	N225	P0602235	M0000188
Malaysia	CKLSE	G0720838	M0161068
Korea	KS11	×	M0161067
Taiwan	TWII	×	M0161063

** Selects the 10-year government bond index due to its more stable fluctuations.*

*** Prediction target.*

This study collected daily closing prices, ten-year bond yields, and exchange rates of various underlying from January 2014 to January 2024. While the period from January 2023 to January 2024 will serve as the verification set for the trading system, the data from January 2014 to January 2023 is further divided to ensure model generalization and facilitate hyperparameter tuning. Specifically, the training data is split in a 6:2:2 ratio: 60% as the training set, used for parameter learning; 20% as the validation set, employed for hyperparameter tuning and model selection to prevent overfitting; and 20% as the test set, used to evaluate the model's final performance during the training phase. The data begins in January 2014 because this marks a significant shift in global economic dynamics, with China's GDP surpassing that of the United States, enhancing China's global economic influence. Additionally, during this period, China's import and export volumes, RMB exchange rates, and other economic indicators reached substantial levels, impacting the global economic landscape. Major events such as Brexit in 2017, the COVID-19 pandemic in 2019, and the Russia-Ukraine conflict in 2022 occurred within this timeframe, providing a diverse set of conditions for model evaluation. While the inclusion of these crisis periods may affect the model's predictability, they offer a comprehensive test of the model's performance under varying economic conditions. The data from January 2023 to January 2024 is considered robust for assessing the trading system, as it allows for a more comprehensive assessment of the system's overall performance, including stability, robustness, and adaptability.

Table 4 displays the codes for the cross-national and cross-asset features collected in this study, excluding indices from the bond dataset with insufficient data.

TABLE 5. Search range of hyperparameters for different models.

Algorithm	Parameter	Value Range
MLP	Batch size	32, 64
	Layer1_neurons	32, 64, 128
	Layer2_neurons	16, 32, 64, 128
	Learning rate	0.001, 0.01, 0.1
LSTM	Batch size	32, 64
	LSTM_layers	1, 2, 4
	Hidden size	64, 128, 256
	Learning rate	0.001, 0.01, 0.1
	Dropout_rate	0.001, 0.01, 0.1
GRU	Batch size	32, 64
	GRU_layers	1, 2, 4
	Hidden size	32, 64, 128
	Learning rate	0.001, 0.01, 0.1
	Dropout_rate	0.001, 0.01, 0.1
Transformer	Batch size	32, 64
	Heads	8, 16
	Embedding dimension	256, 512
	Feed forward neurals	1024, 2048
	Multi-Head attention	64, 128
	Encoding-Decoding layers	8, 16

B. MODELING TECHNIQUES

Upon preparation of the dataset, data preprocessing (detailed in Section III-A) will be conducted to generate the input data for the deep learning model. The training phase significantly influences the model's predictive performance. Establishing a framework for hyperparameter optimization can substantially enhance predictive outcomes [48]. This study utilizes grid search, as delineated in Table 5, to examine hyperparameter ranges across four distinct models. Hyperparameter combinations will be adjusted based on the unique characteristics of each dataset to optimize model prediction capabilities [49].

When training a cross-market model, follow these four steps to identify the optimal structural hyperparameter combination for the four models in Table 5. First, search the hyperparameter range for each model. Second, use the identified hyperparameter group to initialize the model. Third, evaluate the model's predictive performance on the test set and record the MSE. Finally, compile the evaluation results for all hyperparameter groups. Each model undergoes grid search to determine a hyperparameter set for training, with only the top five model structure hyperparameters, ranked by MSE, being considered.

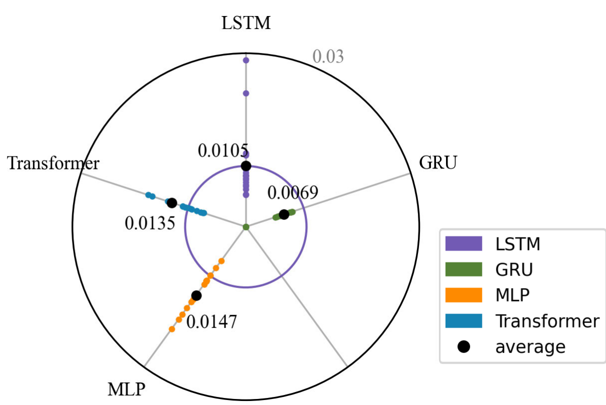
It is important to note that the experiment revealed overfitting in some hyperparameter groups during hyperparameter training rounds. Consequently, the training rounds are fixed at 200. Parameter combinations that fail to converge are excluded based on the validation set MSE visual graphics, retaining only those with satisfactory convergence Appendix B). Repeat these steps for each dataset of different assets to obtain all MSE results. The structural hyperparameter group that appears most frequently in the statistical results is considered the optimal model structure for cross-market datasets of different assets (detailed statistical results are in Appendix C), as illustrated in Table 6.

C. RESULTS

In this section, four models of optimal structures are studied, and MSE evaluations are conducted on the test sets of various investment targets datasets. The deep learning model with the best evaluation results is then selected to construct a trading system, which is evaluating the daily average return rate, risk rate, and Sharpe ratio.

1) MSE

This study investigates the influence of international information on the Chinese stock market by analyzing the prediction performance of different asset datasets within the model. Additionally, it assesses the significance of cross-market data as labels in forecasting the Chinese stock market.



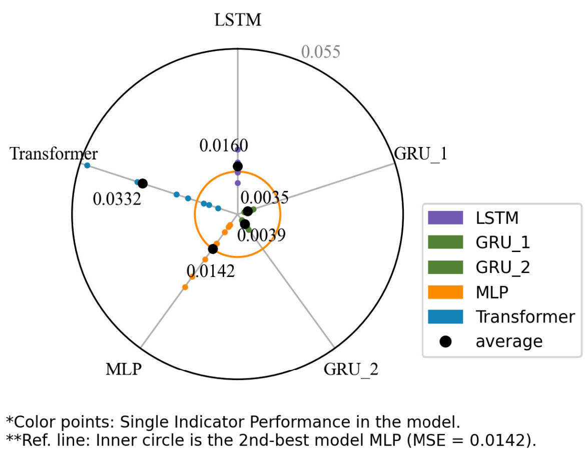
*Color points: Single Indicator Performance in the model.
**Ref. line: Inner circle is the 2nd-best model LSTM (MSE = 0.0105).

FIGURE 3. Mean squared error radar chart for Cross-National Stock Market Data.

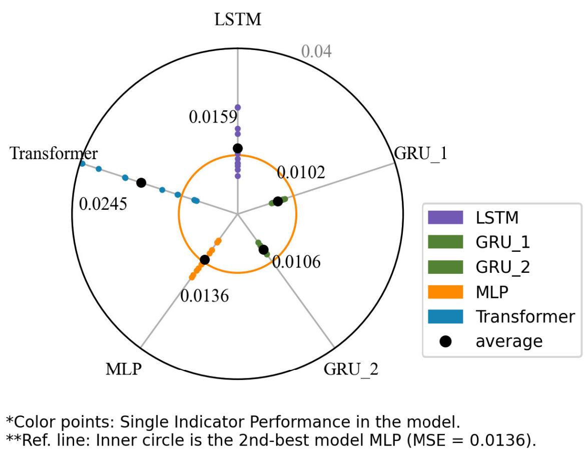
TABLE 6. Optimal structural hyperparameters for four models corresponding to various investment targets.

Algorithm	Stock Market	Bond Market	Currency Rate Market
MLP	Layer1_neurons:128 Layer2_neurons:128	Layer1_neurons:128 Layer2_neurons:64	Layer1_neurons:128 Layer2_neurons:16
LSTM	LSTM_layers:1 Hidden_size:256	LSTM_layers:2 Hidden_size:64	LSTM_layers:1 Hidden_size:256
GRU	GRU_layers:1 Hidden_size:32	GRU_layers:1 *or* GRU_layers:1* Hidden_size:64 Hidden_size:128	GRU_layers:1 *or* GRU_layers:1* Hidden_size:64 Hidden_size:128
Transformer	Heads:8 Embedding dimension:256 Feed forward neurals:1024 Multi-Head attention:64 Encoding-Decoding layers:8	Heads:8 Embedding dimension:256 Feed forward neurals:1024 Multi-Head attention:64 Encoding-Decoding layers:8	Heads:8 Embedding dimension:512 Feed forward neurals:2048 Multi-Head attention:128 Encoding-Decoding layers:8

** The dataset includes two GRU models: GRU_1 with {gru_layers:1, hidden_size:64} and GRU_2 with {gru_layers:1, hidden_size:128}.*



(a) Bond Market Data



(b) Currency Rate Data

FIGURE 4. Mean squared error radar chart for cross-national, cross-asset data.

Figure 3 and 4 present radar charts of MSE tests on cross-market datasets with different labels for four models. The proximity of the black dots to the center indicates the model's predictive capability: the closer to the center, the lower the average MSE, signifying better predictive performance; conversely, the farther from the center, the weaker the predictive ability. The purple circle in Figure 3 marks the MSE value corresponding to a unit length of 0.0105, representing the average MSE of the LSTM model on the cross-national stock dataset, thereby facilitating comparison of MSE performance among the four models.

It is evident that the average MSE of the GRU model is 0.0036 lower than that of the LSTM model, while the average MSEs of both the MLP and Transformer models are higher than that of the LSTM model. Therefore, the GRU model demonstrates the best predictive performance on the cross-national stock datasets.

Similarly, Figure 4 displays the performance of five models, including two GRU models with different structural hyperparameters, on cross-national, cross-asset labeled datasets. Figure 4(a) and 4(b) show radar charts of the MSE tests for bond and exchange rate datasets, respectively. The similarity in performance of the GRU models with different structures in these datasets is due to their optimal hyperparameter settings for cross-national and cross-market data.

For comparison, a circle is drawn using the average MSE value of the second-ranked model, specifically the MLP's average MSE value in the dual-label data. It is evident that the GRU_1 and GRU_2 models outperform the other models. In Figure 4(a), the GRU shows an improvement of 0.01 compared to the MLP, while in Figure 4(b), the GRU model is only 0.003 better than the MLP. This discrepancy can be attributed to the greater stability of the ten-year bond data compared to the exchange rate data, resulting in superior predictive performance by the models.

The results in the Table 7 show that deep learning models significantly outperform the traditional ARIMA model in financial time series forecasting, with GRU achieving the best performance on cross-market datasets. Specifically, the MSE for GRU is 0.0069 on the cross-national stock dataset, 0.0035 on the cross-national cross-asset bond dataset, and 0.0102 on the cross-national cross-asset exchange rate dataset. LSTM performs second best, while Transformer shows average results under the current experimental setup. MLP also demonstrates certain competitiveness. Due to its difficulty in capturing complex temporal features, ARIMA exhibits a significantly higher MSE compared to deep learning models, further confirming the superiority of deep learning methods in financial market forecasting. One of the key factors contributing to ARIMA's lower performance is the limitation of its linear assumption, as it relies on a linear autoregressive structure that fails to capture the nonlinear patterns commonly present in market price fluctuations. Furthermore, although ARIMA addresses non-stationarity through differencing, its treatment remains insufficient. In contrast, deep learning models overcome these limitations through nonlinear activation functions and adaptive feature extraction, enabling robust modeling of complex financial dynamics.

TABLE 7. Table comparison of average MSE between deep learning models and classical models.

model	Stock Market	Bond Market	Currency Rate Market
ARIMA	0.27	0.59	0.57
LSTM	0.0105	0.0160	0.0159
GRU*	0.0069	0.0035	0.0102
MLP	0.0147	0.0142	0.0136
Transformer	0.0135	0.0332	0.0245

** The table shows the best results for the GRU model with different parameters.*

From the results above, it can be seen that the GRU model performs better when applied to cross-market datasets. Specifically, the MSE values for the GRU model are 0.0069 for the cross-national stock dataset, 0.0035 for the cross-national cross-asset bond dataset, and 0.0102 for the cross-national cross-asset exchange rate dataset. Although the MSE improvement in the cross-national cross-asset bond dataset is only 0.0034, it indicates that cross-market information has a positive influence on the prediction of the Chinese stock market. Despite the seemingly small improvement in MSE, it helps enhance the model's ability to accurately predict future trends, which in turn improves returns and reduces risks in actual trading. Therefore, even small MSE improvements may accumulate into significant advantages in long-term trading strategies, especially when considering trading frequency and transaction costs.

2) INVESTMENT PERFORMANCE

From Section IV-C1, it is evident that the GRU model demonstrates superior performance across all three types of cross-market datasets. Consequently, this study integrates algorithms and arbitrage strategies to develop a trading system based on the GRU-Arbitrage rule for investment evaluation. We utilize data from January 2023 to January 2024 as the test set and SZ399001 as the underlying stock for historical backtesting. This evaluation examines performance metrics such as daily average return rate, risk rate, and Sharpe ratio to determine the efficacy of the trading system for cross-market data.

TABLE 8. The daily average returns of different strategies in the test range of three cross-market data sets.

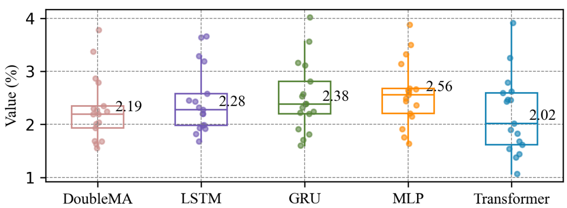
Strategy	Stock Market	Bond Market	Currency Rate Market
Buy and hold	0.78% 0.82%	0.72% 0.73%	0.58% 0.59%
DoubleMA	2.16% 2.29%	2.03% 2.1%	1.9% 2.1%
GRU_Arbitrage_0	2.76% 2.85%	2.96% 2.99%	2.4% 2.44%
GRU_Arbitrage_0.0005	2.64% 2.72%	2.88% 2.91%	2.23% 2.29%
GRU_Arbitrage_0.001	2.51% 2.62%	2.8% 2.84%	2.02% 2.07%
GRU_Arbitrage_0.005	1.68% 1.77%	1.75% 1.81%	0.89% 0.95%

Table 8 presents the investment efficiency of the GRU trading system, with and without transaction costs, after applying five trading rules (see Section III-C1 for details). In this study, the actual transaction cost is considered to be 2.5% of the transaction amount. The results clearly indicate that the four arbitrage strategies based on deep learning algorithms statistically outperform the baseline strategy.

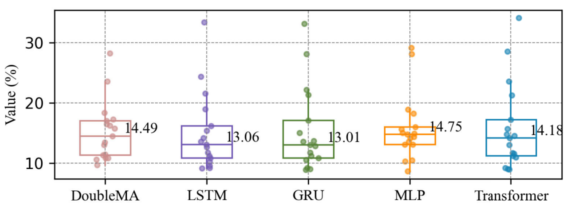
In this study, the dual moving average crossover strategy is used as one of the baseline strategies. This strategy relies on two moving averages with different periods: a short-period moving average (fast MA) to capture short-term price fluctuations and a long-period moving average (slow MA) to identify long-term trends. A buy signal is triggered when the fast MA crosses above the slow MA, while a sell signal is generated when the fast MA crosses below the slow MA. Although the dual moving average strategy performs well in trending markets, its lagging nature makes it prone to frequent trades in ranging markets, leading to higher transaction costs and slippage. Given that the transaction cost in this study is assumed to be 2.5% of the transaction amount, the actual returns of this strategy may be significantly impacted. Moreover, compared to deep learning-based arbitrage strategies, the dual moving average strategy fails to leverage complex market features, resulting in inferior statistical performance.

However, stricter trading rules lead to a decrease in transaction frequency, transaction costs, and returns. This outcome demonstrates that the GRU algorithm effectively leverages cross-market data to accurately reflect the characteristics of the Chinese stock market with minimal error.

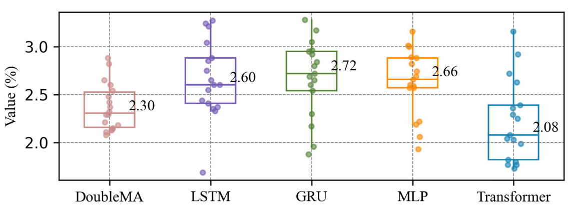
We also replaced the GRU algorithm with the remaining three deep learning algorithms in the GRU_Arbitrage_0 strategy and compared the investment performance of the GRU-Arbitrage trading system against these alternatives.



(a) Daily Average Returns



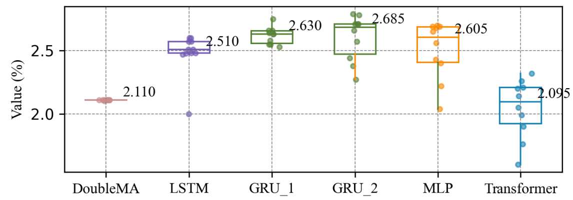
(b) Risk Measures



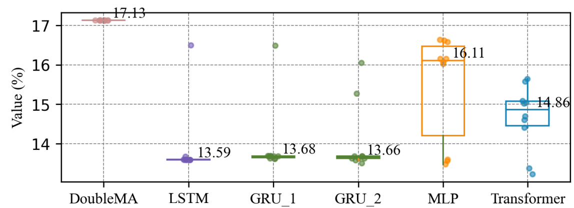
(c) Sharpe Ratios

FIGURE 5. Box plots of investment performance variation using Stock Market Dataset as labels across five algorithmic trading systems.

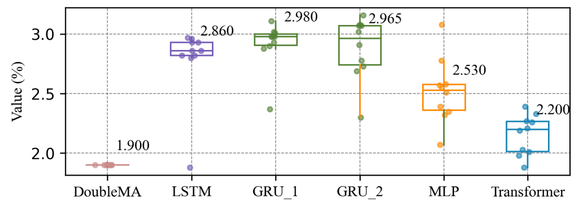
Figure 5 presents box plots of daily average returns (Figure 5a), risk (Figure 5b), and Sharpe ratio (Figure 5c) after backtesting five algorithmic trading systems on the Chinese stock index, using cross-national stock market index data as labels. The comparison also includes a more advanced double moving average (DoubleMA) strategy as a benchmark.



(a) Daily Average Returns



(b) Risk Measures



(c) Sharpe Ratios

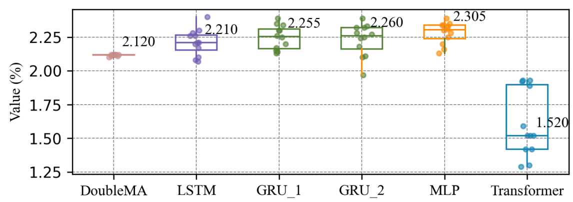
FIGURE 6. Box plots of investment performance variation using Bond Market Dataset as labels across five algorithmic trading systems.

From Figure 5a, it is evident that the MLP achieves the highest daily average return of 2.56%, followed by the GRU with an average return of approximately 2.38%. However, when considering risk indicators, the GRU demonstrates superior stability, maintaining a risk level around 13.01%. This is followed by the LSTM, with a risk of 13.06%, slightly higher than that of the GRU. The remaining three algorithmic trading systems exhibit risks exceeding 14%, indicating poorer performance.

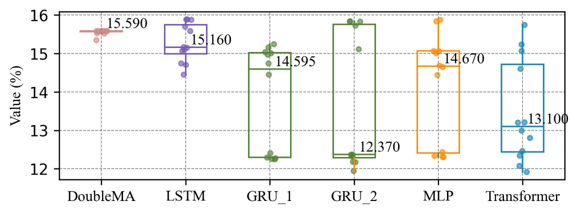
Overall, the GRU-based trading system demonstrates the best investment performance, which is also reflected in the Sharpe ratio. As shown in Figure 5c, the Sharpe ratio of the GRU-based system reaches 2.72, surpassing that of the MLP at 2.66, followed by the LSTM at 2.60, the DoubleMA at 2.30, and the Transformer at 2.08. Therefore, for investment strategies using cross-national stock index data as a single feature label, the GRU-based trading system exhibits the best performance, achieving relatively high returns while maintaining the lowest strategy risk.

The GRU algorithm-based trading system also performs well in terms of the Sharpe ratio, as shown in Figure 5(c). The Sharpe ratio for the GRU-based system is 2.72, surpassing the MLP's 2.66, the LSTM's 2.60, and the Transformer's 2.08. Therefore, using cross-national stock index data as a single feature label, the GRU algorithm-based trading system demonstrates the best overall investment performance, achieving relatively high returns while maintaining the lowest risk among the evaluated strategies.

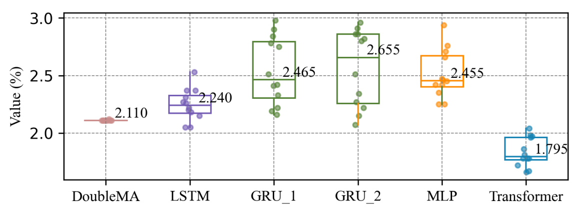
Furthermore, Figures 6 and 7 illustrate the investment performance of trading systems utilizing cross-national and cross-asset index data as labels, comparing the double moving average algorithm with five deep learning algorithms, including two GRU models with different structural hyperparameters.



(a) Daily Average Returns



(b) Risk Measures



(c) Sharpe Ratios

FIGURE 7. Box plots of investment performance variation using Currency Rate Market Dataset as labels across five algorithmic trading systems.

In Figure 6(a) and 7(a), it is evident that the LSTM, GRU, and MLP models all achieve strong daily average returns with cross-national, cross-asset data. However, when evaluating risk, Figure 6(b) reveals that only the LSTM and GRU maintain relatively low volatility, around 13.6%, for cross-national, cross-asset bond data. Figure 7(b) shows that, for exchange rate data, only the GRU trading system exhibits a risk below 13%, with other models performing less favorably. This difference can be attributed to data stability, as ten-year bond data is more stable than exchange rate data, resulting in better performance for some algorithmic trading systems.

Figure 6(c) and 7(c) demonstrate that the GRU consistently achieves the highest Sharpe ratio compared to the other three deep learning trading systems. In bond data, the GRU's Sharpe ratio exceeds that of the LSTM by 0.15, and in exchange rate data, it surpasses the MLP's Sharpe ratio by 0.2. This aligns with the observation that the average Sharpe ratio for bond data is generally higher than that for exchange rate data. The double moving average strategy exhibits high risk and relatively low returns when applied to cross-national and cross-asset data.

However, as shown in the figures, the predicted values obtained for each asset are relatively similar. This can be explained by the fact that the strategy is best suited for markets with clear trends and high volatility. Since the fluctuation patterns of the foreign exchange and bond markets differ from those of the stock index market, directly using their data to predict stock indices may lead to trend mismatches, resulting in distorted trading signals and increased risk.

Overall, the GRU-Arbitrage algorithm-based trading system demonstrates superior performance across cross-national, cross-asset datasets.

V. CONCLUSION

This study primarily examines the fluctuations of China's stock market through dynamic information from multiple markets, utilizing both cross-national stock data and cross-national cross-asset data. Our empirical findings reveal that data on similar assets across different countries and data on various asset types across countries significantly impact the Chinese stock market. This indicates a strong correlation between the Chinese stock market and international market fluctuations.

Our research verifies that cross-market data can enhance prediction accuracy, increase returns, and reduce risks in investments within the Chinese stock market. We identified the optimal structural hyperparameters for four models that better fit cross-market data and developed a reproducible framework. The results demonstrate that the GRU algorithm performs effectively when predicting the Chinese stock index using various target datasets as labels. By integrating the optimal structural model with the arbitrage strategy, we constructed a GRU-Arbitrage algorithm trading system. The final results show that the GRU_Arbitrage strategy outperforms the baseline strategy under various conditions. Notably, the GRU_Arbitrage_0 strategy, which directly compares the closing price with the arbitrage rule, yields an average daily return rate approximately 2% higher than the baseline strategy.

However, this study also has some limitations. First, the interpretability of deep learning models remains a challenge. Models like GRU are often regarded as ''black-box'' models, which makes it difficult to explain how predictions are made. This lack of transparency may limit practitioners' trust in the model. Secondly, this study focuses solely on a single stock index (SZ399001) from the Chinese stock market, which may limit the generalizability of the results. Applying this model to other markets or stock indices might yield different outcomes.

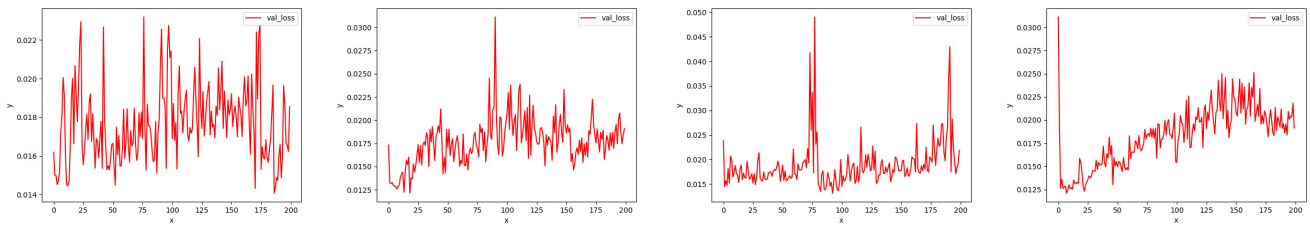


FIGURE 8. Display of converged MSE results.

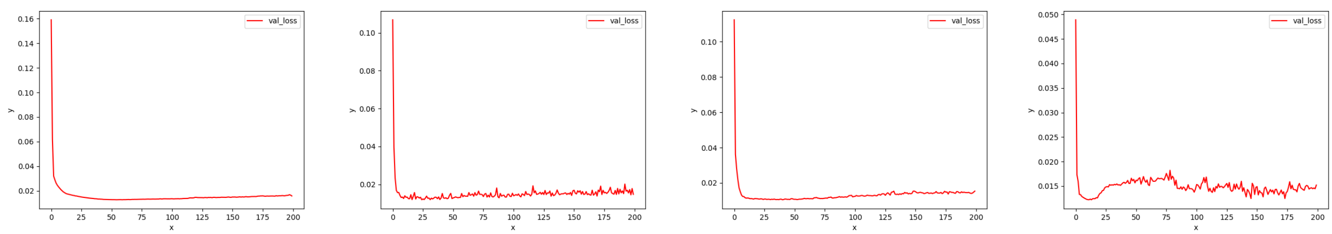


FIGURE 9. Display of non-convergent MSE results.

TABLE 9. Preliminary validation MSE results for T_w with different model fixation parameters and baseline metrics.

Dataset	Model	T_w=15	T_w=20	T_w=25	T_w=30	T_w=35	T_w=40	T_w=45	T_w=50
Stock	GRU	0.0077	0.0058	0.0339	0.0096	0.0076	0.0068	0.0046	0.0221
	LSTM	0.0105	0.0080	0.0323	0.0105	0.0257	0.0083	0.0139	0.0739
	MLP	0.0343	0.0129	0.0305	0.0305	0.0178	0.0133	0.0083	0.0255
	Transformer	0.0599	0.0072	0.0382	0.0186	0.0648	0.0213	0.0151	0.0742
Bond	GRU	0.0085	0.0059	0.0080	0.0057	0.0072	0.0192	0.0067	0.0048
	LSTM	0.0056	0.0040	0.0093	0.0085	0.0096	0.0076	0.0165	0.0049
	MLP	0.0148	0.0063	0.0129	0.0167	0.0070	0.0156	0.0149	0.0267
	Transformer	0.0744	0.0130	0.0144	0.0158	0.0088	0.0376	0.0198	0.0144
Currency Rate	GRU	0.0089	0.0079	0.0085	0.0113	0.0089	0.0083	0.0125	0.0149
	LSTM	0.0116	0.0118	0.0128	0.0170	0.0047	0.0113	0.0134	0.0162
	MLP	0.0125	0.0102	0.0077	0.0163	0.0172	0.0132	0.0116	0.0113
	Transformer	0.4545	0.0179	0.0908	0.0158	0.1329	0.0255	0.0271	0.0343

Despite these limitations, the findings of this study still validate the effectiveness of the adopted strategies and underscore the potential of cross-market data forecasting in predicting the Chinese stock market. For practitioners, incorporating cross-national and cross-asset data can enhance prediction accuracy and investment outcomes, particularly in complex market environments. The study also offers theoretical insights into applying cross-market data for stock index prediction, contributing to the optimization of investment strategies. Future research could explore other markets and asset classes, such as futures, oil, and cryptocurrencies, to test the universality of this approach. Additionally, further exploration of alternative time series feature extraction methods would be a valuable direction for future research.

APPENDIX A
PARTIAL VISUALIZATION OF DIFFERENT STRUCTURAL MODELS IN THE DATASET

See Figures 8 and 9.

APPENDIX B

See Table 9.

APPENDIX C

See Table 10.

TABLE 10. The optimal number of times that hyperparameters of model structure occur across-market datasets.

Algorithm	Stock Market		Bond Market		Currency Market	
	Structural Hyperparameters	T	Structural Hyperparameters	T	Structural Hyperparameters	T
GRU	GRU_layers:1; Hidden_size:32	29	GRU_layers:1; Hidden_size:64	10	GRU_layers:1; Hidden_size:64	15
	GRU_layers:2; Hidden_size:32	26	GRU_layers:1; Hidden_size:128	10	GRU_layers:1; Hidden_size:128	15
	GRU_layers:2; Hidden_size:64	13	GRU_layers:2; Hidden_size:128	6	GRU_layers:2; Hidden_size:128	10
	GRU_layers:2; Hidden_size:128	9	GRU_layers:4; Hidden_size:32	5	GRU_layers:1; Hidden_size:32	7
	GRU_layers:1; Hidden_size:64	8	GRU_layers:4; Hidden_size:64	5	GRU_layers:2; Hidden_size:32	6
LSTM	LSTM_layers:1; Hidden_size:256	33	LSTM_layers:2; Hidden_size:64	10	LSTM_layers:1; Hidden_size:256	15
	LSTM_layers:1; Hidden_size:128	15	LSTM_layers:1; Hidden_size:256	9	LSTM_layers:1; Hidden_size:128	11
	LSTM_layers:2; Hidden_size:128	13	LSTM_layers:1; Hidden_size:128	8	LSTM_layers:1; Hidden_size:64	10
	LSTM_layers:1; Hidden_size:64	11	LSTM_layers:2; Hidden_size:128	6	LSTM_layers:2; Hidden_size:64	9
	LSTM_layers:2; Hidden_size:64	6	LSTM_layers:4; Hidden_size:64	5	LSTM_layers:2; Hidden_size:128	8
MLP	Layer1_neurons:128; Layer2_neurons:128	19	Layer1_neurons:128; Layer2_neurons:64	12	Layer1_neurons:128; Layer2_neurons:16	14
	Layer1_neurons:128; Layer2_neurons:16	18	Layer1_neurons:128; Layer2_neurons:32	10	Layer1_neurons:128; Layer2_neurons:128	13
	Layer1_neurons:128; Layer2_neurons:32	17	Layer1_neurons:64; Layer2_neurons:16	8	Layer1_neurons:128; Layer2_neurons:64	10
	Layer1_neurons:128; Layer2_neurons:64	17	Layer1_neurons:64; Layer2_neurons:128	8	Layer1_neurons:128; Layer2_neurons:16	9
	Layer1_neurons:64; Layer2_neurons:128	5	Layer1_neurons:128; Layer2_neurons:16	6	Layer1_neurons:128; Layer2_neurons:32	7
Transformer	Heads:8; Embedding_dimension:256; Feed_forward_neurals:1024; Multi_Head_attention:64; Encoding_Decoding_layers:8	9	Heads:8; Embedding_dimension:256; Feed_forward_neurals:1024; Multi_Head_attention:64; Encoding_Decoding_layers:8	6	Heads:8; Embedding_dimension:512; Feed_forward_neurals:2048; Multi_Head_attention:128; Encoding_Decoding_layers:8	8

TABLE 10. ***(Continued.)*** **The optimal number of times that hyperparameters of model structure occur across-market datasets.**

Heads:8; Embedding_dimension:256; Feed_forward_neurals:2048; Multi_Head_attention:64; Encoding_Decoding_layers:8	7	Heads:8; Embedding_dimension:256; Feed_forward_neurals:2048; Multi_Head_attention:64; Encoding_Decoding_layers:16	4	Heads:8; Embedding_dimension:512; Feed_forward_neurals:2048; Multi_Head_attention:64; Encoding_Decoding_layers:8	7
Heads:8; Embedding_dimension:256; Feed_forward_neurals:2048; Multi_Head_attention:128; Encoding_Decoding_layers:8	7	Heads:16; Embedding_dimension:256; Feed_forward_neurals:1024; Multi_Head_attention:128; Encoding_Decoding_layers:16	4	Heads:16; Embedding_dimension:512 ; Feed_forward_neurals:2048; Multi_Head_attention:64; Encoding_Decoding_layers:8	7
Heads:8; Embedding_dimension:512; Feed_forward_neurals:1024; Multi_Head_attention:64; Encoding_Decoding_layers:8	6	Heads:16; Embedding_dimension:512; Feed_forward_neurals:2048; Multi_Head_attention:64; Encoding_Decoding_layers:8	4	Heads:8; Embedding_dimension:512; Feed_forward_neurals:1024; Multi_Head_attention:64; Encoding_Decoding_layers:8	6

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